

c)Proportional scheduling

code:

```
#include<stdio.h>
#include <stdlib.h>
#include <time.h>
typedef struct
{
char name[5]; int tickets; }
Process;
int main() {
printf("usn:1WA24CS235\n");
printf("name:Riya gupta\n");
int n, total_tickets = 0; float total_T =0.0;
printf("Enter the number of Processes: ");
scanf("%d", &n);
Process p[n];
srand(time(NULL));
for (int i = 0; i < n; i++)
{ printf("\nProcess %d:\n", i + 1);
sprintf(p[i].name, "P%d", i + 1);
printf("Tickets: ");
scanf("%d", &p[i].tickets);
total_tickets += p[i].tickets;
total_T +=p[i].tickets;
}
printf("\n--- Proportional Share Scheduling ---\n");
printf("Enter the Time Period for scheduling: ");
int m;

scanf("%d",&m);
for (int i = 0; i < m; i++)
{
int winning_ticket = rand() % total_tickets + 1;
int accumulated_tickets = 0;
int winner_index;
for (int j = 0; j < n; j++)
{
accumulated_tickets += p[j].tickets;
if (winning_ticket <= accumulated_tickets)
```

```
{  
winner_index = j; break;  
}  
}  
printf("Tickets picked: %d, Winner: %s\n", winning_ticket,  
p[winner_index].name);  
}  
for (int i = 0; i < n; i++)  
{  
printf("\nThe Process: %s gets %0.2f%% of Processor Time.\n", p[i].name,  
((p[i].tickets / total_T) * 100));  
  
} return 0; }
```

output:

Output

Clear

```
usn:1WA24CS235
name:Riya gupta
Enter the number of Processes: 3

Process 1:
Tickets: 10

Process 2:
Tickets: 20

Process 3:
Tickets: 30

--- Proportional Share Scheduling ---
Enter the Time Period for scheduling: 5
Tickets picked: 18, Winner: P2
Tickets picked: 9, Winner: P1
Tickets picked: 10, Winner: P1
Tickets picked: 51, Winner: P3
Tickets picked: 51, Winner: P3

The Process: P1 gets 16.67% of Processor Time.

The Process: P2 gets 33.33% of Processor Time.

The Process: P3 gets 50.00% of Processor Time.
```